LOAN APPROVAL PREDICTION

Machine Learning Internship Project Report

SUBMISSION DETAILS	
Student Name	BHUMIKA SRIVASTAVA
Course	Bachelor of Computer Applications (BCA)
College	NETAJI SUBHAS UNIVERSITY
Internship Organization	InternSpark
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Project Title	Loan Approval Prediction

Submitted as part of the internship task requirements.

EXECUTIVE SUMMARY

This project focuses on developing a machine learning model to predict loan approval decisions based on applicant information. The purpose is to understand how data-driven techniques can support financial institutions in evaluating loan applications.

The project includes data cleaning, missing-value treatment, categorical encoding, feature scaling and class imbalance handling. Multiple classification algorithms are compared using standard evaluation metrics.

The results help identify a suitable model and demonstrate how prediction thresholds can influence financial decision-making.

1. Introduction

Loan approval is an important activity for banks and financial institutions. Before approving a loan, lenders evaluate information such as an applicant's income, employment status, credit history and other financial details.

Machine learning can analyse historical application data and identify patterns associated with loan approval. This project applies supervised machine learning to predict loan approval outcomes.

2. Project Objectives

• Understand and analyse the loan approval dataset.

• Identify and handle missing values and duplicate records.

• Convert categorical information into numerical features.

• Scale numerical features where required.

• Address class imbalance using SMOTE.

• Train and compare different classification algorithms.

• Evaluate model performance using appropriate metrics.

• Interpret results from a business perspective.

3. Dataset Description

The dataset used in this project is the Loan Approval Prediction dataset available on Kaggle. It contains information about loan applicants and their application outcomes.

Important features include applicant income, co-applicant income, education, employment status, loan amount, credit history and property area. The target variable represents whether the loan was approved or rejected.

Dataset source: Kaggle - Loan Approval Prediction

4. Data Preprocessing

Data preprocessing prepares the raw dataset for machine learning. Missing numerical values were handled using median imputation, while missing categorical values were replaced using the most frequent category.

Categorical features were transformed using One-Hot Encoding. Numerical features were scaled using StandardScaler. Duplicate records and unnecessary identifier columns were also handled.

5. Handling Class Imbalance

Class imbalance occurs when one target category has significantly more records than another. This can affect a model's ability to recognise the minority category.

SMOTE (Synthetic Minority Over-sampling Technique) was used to generate synthetic examples of the minority class in the training data. This helps the model learn from both categories. SMOTE was applied within the training pipeline to prevent data leakage.

6. Machine Learning Models

Logistic Regression: A classification algorithm that estimates the probability of a loan approval outcome. It provides a simple and interpretable baseline.

Decision Tree: A model that makes predictions through a series of feature-based decisions. It can capture non-linear relationships.

Random Forest: An ensemble learning algorithm that combines multiple decision trees to produce predictions. It can provide more stable results than a single decision tree.

7. Model Evaluation and Comparison

The models were evaluated using Precision, Recall, F1-score and ROC-AUC. These metrics provide different ways to understand the quality of classification predictions.

Precision: Measures the proportion of predicted positive cases that are actually positive.

Recall: Measures the proportion of actual positive cases correctly identified by the model.

F1-score: Combines precision and recall into a single measure.

ROC-AUC: Measures how well a model distinguishes between the two target classes across different thresholds.

Model	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	Enter	Enter	Enter	Enter
Decision Tree	Enter	Enter	Enter	Enter
Random Forest	Enter	Enter	Enter	Enter

Best-performing model: Enter the best-performing model

8. Classification Threshold

A classification threshold determines how predicted probabilities are converted into final class predictions. A default threshold of 0.50 can be compared with a threshold selected using validation data.

Selected threshold: Enter the selected threshold

The final threshold should balance the risks of incorrectly approving and incorrectly rejecting applications. It should also reflect the institution's lending policy.

9. Business Interpretation

The model can support the initial screening of loan applications. However, incorrect predictions can have financial and customer-related consequences.

A false approval may expose a lender to repayment risk, while a false rejection may prevent a creditworthy applicant from receiving a loan. Therefore, both precision and recall are important when evaluating the model.

10. Deployment Recommendations

The model should initially be used as a decision-support system rather than an automatic loan approval authority. Before deployment, it should be tested on recent data, checked for unfair bias, monitored for performance changes and reviewed by qualified human decision-makers.

11. Conclusion

This project demonstrates the application of supervised machine learning to loan approval prediction. It covers data preprocessing, class imbalance handling, model comparison and performance evaluation.

The project highlights how machine learning can support financial decision-making while also showing the importance of model evaluation, threshold selection, fairness and human oversight.

12. Project Links and References

GitHub Repository: Open GitHub Project

Jupyter Notebook: Open Project Notebook

Dataset: Open Kaggle Dataset